



Web Development Lab

CSS Positioning — Modal Pop-Up Practical Session Guide

17 Guided Steps • Beginner–Intermediate • Estimated Time: 50 mins

Lab Overview

In this session you will build a **modern Modal Pop-Up Component** using **CSS Positioning**.

You will learn how to:

- Use **position: fixed**
- Center elements with **transform**
- Control layering using **z-index**
- Build overlays and floating UI components
- Style forms inside a modal
- Hide and show elements using utility classes

This lab introduces one of the most important real-world UI patterns used in:

- Dashboards
- Authentication systems
- Alert systems
- Project management applications

Each step builds progressively so students can visually understand how CSS positioning works.

Note: This project uses:

- HTML
- CSS
- External CSS file (`modal.css`)

Prerequisites

- Basic HTML knowledge
- Understanding of Flexbox
- Completed previous CSS labs 

1. Create the HTML File Structure

Objective: Set up the modal project structure.

🕒 5 min

Background

- Every web project begins with a structured HTML document.

```
◦ <!DOCTYPE html>
```

- Tells the browser this is an HTML5 document

```
◦ <meta name="viewport">
```

- Makes the page responsive on mobile devices

```
◦ <link rel="stylesheet" href="modal.css">
```

- Connects the external CSS file

Instructions

1. Create:

- `index.html`
- `modal.css`

2. Add the HTML boilerplate

Code Snippet

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Modal Pop CSS Design</title>
  <link rel="stylesheet" href="modal.css">
</head>
<body>

</body>
</html>
```



Tip: External CSS keeps code cleaner and scalable.

2. Create the Page Header

Objective: Build the introduction section.

🕒 5 min

Background

The header introduces the page content and contains the modal trigger button.

Instructions

1. Add the header HTML

Code Snippet

```
<header class="header">
  <p>Modal Pop-Up</p>
  <small>
    A modern modal component with overlay, animations, and form handling
  </small>

  <button class="open-modal-btn">Open Modal</button>
</header>
```



Tip: Modal buttons are called “Triggers”.

3. Style the Page Background

Objective: Create a dark UI theme.

🕒 5 min

Background

- `background-color: #020618;`
 - *Applies dark background*
- `margin: 50px 213.5px;`
 - *Adds outer spacing around the page*

Instructions

1. Style the body

Code Snippet

```
body{  
  background-color: #020618;  
  margin:50px 213.5px ;  
}
```

💡 **Tip:** Dark UI themes are common in dashboards and modals.

4. Convert Header into a Flex Container

Objective: Stack content vertically.

🕒 5 min

Background

- **Display:flex;**
 - Turns header into flex container
- **flex-direction: column;**
 - Stacks children vertically
- **gap: 0px;**
 - Controls spacing between items

Instructions

1. Style `.header`

Code Snippet

```
.header{  
  display:flex;  
  flex-direction: column;  
  gap:0px;  
}
```



Tip: Flexbox is useful even for simple vertical layouts.

5. Style the Header Text

Objective: Improve typography hierarchy.

🕒 6 min

Background

- **font-size: 30px;**
 - Controls text size
- font-weight: bold;
 - **Makes heading stand out**
- margin-bottom: 8px;
 - **Creates spacing below heading**

Instructions

1. Style `.header p`

Code Snippet

```
.header p{  
  font-size: 30px;  
  color: #fff;  
  font-weight: bold;  
  line-height: 36px;  
  margin-bottom: 8px;  
}
```



Tip: Typography creates visual hierarchy.

6. Style the Trigger Button

Objective: Create the modal opening button.

🕒 6 min

Background

- **background-color: #4F39F6;**
 - Sets primary button color
- **border-radius: 10px;**
 - Rounds button corners
- **border: 1px solid #4F39F6;**
 - Adds matching border

Instructions

1. Style `.header button`

Code Snippet

```
.header button{
  background-color: #4F39F6;
  color: #fff;
  width: 137.8px;
  height: 48px;
  border-radius: 10px;
  margin-top: 32px;
  margin-bottom: 48px;
  border: 1px solid #4F39F6;
}
```



Tip: CTA buttons should visually stand out.

7. Add Hover Effects to the Button

Objective: Improve user interaction feedback.

🕒 4 min

Background

- **:hover**
 - Triggers styles during hover
- **background-color: #2209e6;**
 - Changes button color

Instructions

1. Add hover styling

Code Snippet

```
.header button:hover{  
  opacity:1;  
  background-color: #2209e6;  
}
```

💡 **Tip:** Hover states improve UX feedback.

8. Create the CSS Concepts Card

Objective: Build an informational card section.

🕒 6 min

Background

- **background-color: #0F172B;**
 - Creates dark card appearance
- **border-radius: 10px;**
 - Rounds card corners
- **padding: 25px 25px;**
 - Adds internal spacing

Instructions

1. Style `.css-concepts-card`

Code Snippet

```
.css-concepts-card{  
  background-color: #0F172B;  
  max-width: 896px;  
  max-height: 254px;  
  padding: 25px 25px;  
  border-radius: 10px;  
}
```



Tip: Cards group related information visually.

9. Build the Modal Structure

Objective: Create the modal container.

🕒 6 min

Background

The modal contains:

- Header
- Form
- Action buttons

Instructions

1. Add the modal HTML

Code Snippet

```
<div class="modal-form hidden">

  <header>
    <h2>Create New Project</h2>
    <small>Enter project details to get started</small>
  </header>

  <form>
    <label>Project Name</label>
    <input type="text" placeholder="My Awesome Project">

    <label>Description</label>
    <textarea placeholder="Describe your project.."></textarea>

    <label>Category</label>
    <input type="text">
  </form>

  <div class="button-div">
    <button class="cancel-btn">Cancel</button>
    <button class="create-btn">Create Project</button>
  </div>

</div>
```



Tip: Modals are floating UI containers.

10. Position the Modal Using Fixed Positioning

Objective: Center the modal on screen.

🕒 8 min

Background

This is the **MOST IMPORTANT** section of the lab.

- **position: fixed;**
 - Locks modal relative to viewport
- **top: 50%;**
- **left: 50%;**
 - Moves modal midpoint to screen center
 - `transform: translate(-50%, -50%);`
 - Corrects positioning by shifting element back

Instructions

1. Style `.modal-form`

Code Snippet

```
.modal-form {  
  background-color: #1D293D;  
  position: fixed;  
  top: 50%;  
  left: 50%;  
  transform: translate(-50%, -50%);  
}
```



Tip: This is the standard technique for centering modals.

11. Understand Z-Index Layering

Objective: Place modal above page content.

🕒 5 min

Background

- **z-index: 100;**
 - Controls stacking order
- **Higher z-index:**
 - Appears above lower elements

Instructions

1. Add z-index

Code Snippet

```
.modal-form{  
  z-index: 100;  
}
```



Tip: Modals must appear above everything else.

12. Style Form Inputs

Objective: Design modal form fields.

🕒 6 min

Background

- **width: 100%;**
 - Makes inputs responsive
- **border-radius: 10px;**
 - Rounds corners
- **background-color: #0F172B;**
 - Creates dark input style

Instructions

1. Style inputs and textarea

Code Snippet

```
.modal-form input,  
.modal-form textarea{  
  background-color: #0F172B;  
  width: 100%;  
  border-radius: 10px;  
  border: 1px solid #0F172B;  
}
```



Tip: Consistent form styling improves usability.

13. Style Action Buttons

Objective: Create modal action controls.

🕒 5 min

Background

- **display: flex;**
 - Aligns buttons horizontally
- **justify-content: center;**
 - Centers buttons

Instructions

1. Style `.button-div`

Code Snippet

```
.button-div{
  display: flex;
  gap: 16px;
  margin-top: 24px;
  align-items: center;
  justify-content: center;
}
```



Tip: Button grouping improves UX clarity.

14. Hide and Show the Modal

Objective: Understand visibility toggling.

🕒 5 min

Background

- **Display:none;**
 - Completely hides element
- **!important**
 - Forces CSS priority

Instructions

1. Add `.hidden` utility class

Code Snippet

```
.hidden{  
    display:none !important;  
}
```



Tip: Utility classes simplify state management.

15. Connect JavaScript to the Modal

Objective: Select and control modal elements using JavaScript.

🕒 6 min

Background

JavaScript allows us to make the modal interactive.

- **document.querySelector()**
 - Selects an HTML element using its class or selector
- **const modal**
 - Stores the modal element in a variable
- **const openModalBtn**
 - Stores the button that opens the modal
- **const closeModalBtn**
 - Stores the button that closes the modal

All these are variables declared by you, you can give them any name, but they should match what they are being used for.

Instructions

1. Create a file named `modal.js`
2. Add the JavaScript below

Code Snippet

```
// Grab the Modal Content
const modal = document.querySelector(".modal-form")

// Grab the button that will open the Modal
const openModalBtn = document.querySelector(".open-modal-btn")

// Grab the button that will close the modal
const closeModalBtn = document.querySelector(".cancel-btn")
```

💡 Tip: `querySelector()` is one of the most commonly used DOM selection methods in JavaScript.

16. Open and Close the Modal with Events

Objective: Use event listeners to control modal visibility.

🕒 8 min

Background

JavaScript events allow us to respond to user interactions.

- **`addEventListener('click')`**
 - Listens for button clicks
- **`modal.classList.remove('hidden')`**
- Removes the `hidden` class
 - This makes the modal visible
- **`modal.classList.add('hidden')`**
 - Adds the `hidden` class back

This hides the modal

Instructions

1. Add the event listeners below your variables

Code Snippet

```
// Listen to when the open Modal Button is clicked
openModalBtn.addEventListener('click', () =>{
    modal.classList.remove('hidden')
})

// Listen to when the Cancel Button is also clicked
closeModalBtn.addEventListener('click', () =>{
    modal.classList.add('hidden');
})
```

💡 **Tip:** This is called “DOM Manipulation” – dynamically changing HTML using JavaScript.

17. Final Review and Testing

Objective: Validate the complete modal interaction system.

🕒 8 min

Instructions

1. Open the browser
2. Click “Open Modal”
3. Confirm the modal appears
4. Click “Cancel”
5. Confirm the modal disappears
6. Inspect layout and positioning



Final Check

- Modal opens correctly
- Modal closes correctly
- Positioning remains centered
- JavaScript events working properly
- No console errors detected